ECOFOG:

A NOVEL COMPOSITE MESH FOR PASSIVE WATER COLLECTION FROM FOG AND HIGH RELATIVE-HUMIDITY ENVIRONMENTS

Introduction

- Over **2 billion people** face water scarcity worldwide. **[1]**
- This water insecurity crisis is becoming increasingly adverse, requiring the emergence of alternative sources of water. **Fog Harvesting** is a promising solution where specialized systems capture water droplets from air and collect them into containers. **[2]**
- **However**, in recent years, such mesh-based systems have been deployed in **only some** of the suitable regions, and **Raschel Mesh** is one of the only solution that has witnessed wide-scale deployment. **[3]**
- Though scalable, Raschel Mesh has a **low efficiency. [4]**



Fig. 1: Map showing global locations where fog harvesting has been successfully implemented & demonstrates potential for future development. **[2]**

Background

Review of Existing Systems

FIELD-IMPLEMENTED



Fig. 2: CloudFisher Raschel Mesh installed in Morocco. **[5]**

LAB-SCALE (NOT-DEPLOYED)

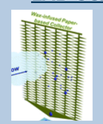


Fig. 3: Biomimetics (Cactus Kirigami) **[6]**



Fig. 4: Janus Meshes [7]

Limitations of Existing Systems



Low Capture

Source: Canva



Complex Fabrication

Source: Canva



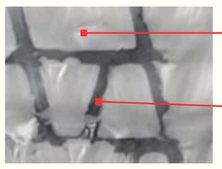
Toxic, Non-Biodegradable Materials

Source: Canva

The above-mentioned problems highlight the need for a more scalable material for moisture and fog harvesting.

Hypothesis

- Utilising electrospun Polyvinyl alcohol (PVA) and Polyethylene oxide (PEO) nanofibers, embedded with Titanium Dioxide (TiO_2) nanoparticles (fillers) can enable capture of high amounts of water droplets.
- After capture, faster drainage of water droplets by utilising a hydrophobic Polylactic Acid (PLA) backbone can ultimately improve the water collection rate through widely-used, non-toxic materials and scalable techniques.



Hydrophilic Matrix for nucleation and coalescense (capture)

Hydrophobic Backbone for droplet detachment and drainage

Fig.5: Crosslinked Electrospun Mesh Close-up. *Photo Taken by Finalists, 2026.*

Materials

- **Polylactic acid (PLA) -** Hydrophobic Polymer, readily used in 3D Printing.
- **Polyvinyl alcohol (PVA) -** Hydrophilic Polymer.
- **Polyethylene oxide (PEO) -** Increases solution rheology and electrospinning jet stability.
- **Titanium Dioxide (TiO_2) -** Provides nucleation sites to enhance droplet capture.
- **Colloidal Silica (SiO_2) -** Its Si-OH bonds form hydrogen bonds with -OH/-NH in the polymer.
- **Glutaraldehyde ($C_5H_8O_2$) -** Its aldehyde groups form bonds with hydroxyl groups in the PVA, creating acetal linkages to increase aqueous stability.

Methodology

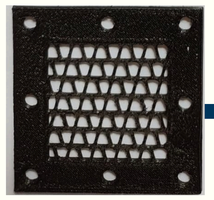


Fig. 6: Pure PLA Backbone. *Photo taken by finalists, 2025.*

A PLA Raschel-style mesh was **3D printed** and optimised to a shade coefficient of 38%.

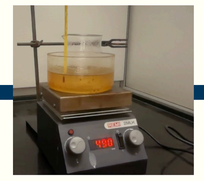


Fig. 7: Solution preparation for electrospinning process. *Photo taken by finalists, 2026.*

Composite Solution (18wt%): PVA (90%, 90 °C oil bath, 250 rpm/ 2h); PEO (10%, 60 °C) in 100 cm^3 water; TiO_2 and SiO_2 nanofillers, sonicated for 15 minutes in distilled water.

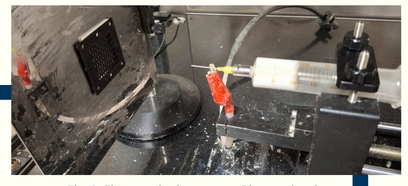


Fig. 8: Electrospinning set-up. *Photo taken by finalists, 2026.*

- **Needle Setup:** Single-needle (18G)
- **Collector:** Static plate
- **Time:** 30 minutes

The electrospinning solution was prepared with varying concentrations of TiO_2 and SiO_2 (1 wt% - 3 wt%) nanofillers and the base PVA/PEO matrix (18 wt%). Based on the material characterization results and the water collection rate, the optimised nanofiller concentration was found. The presented results and photos correspond to this optimum concentration.

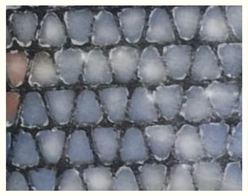


Fig. 10: Final mesh post crosslinking. *Photo taken by Finalists, 2026.*



Fig. 9: Crosslinking of mesh by glutaraldehyde vapours. *Photo taken by Finalists, 2026.*

- Exposed to 2.5% of glutaraldehyde solution vapour for one hour
- Washed with acetone to remove residual glutaraldehyde

Material Characterization and Testing

Water Collection Rate Testing

CONTROL VARIABLES

- Fog Generation Rate
- Exposure Time (4 hours)
- Airflow / Wind Speed (3.0 m/s)
- Chamber Conditions (90±1% RH, 20±2°C)
- Mesh Orientation and Position

Material Characterization

- **FESEM** - Nanofiber morphology
- **Contact angle** - Hydrophilicity and Hydrophobicity of components
- **FTIR** - Successful incorporation of nanoparticles
- **XRD** - TiO_2 retention

Field Testing: EcoFog, PLA backbone, and Raschel mesh (identical area, mounting, orientation) were deployed simultaneously outdoors over 5 days during natural high-humidity events (RH 90–95%, Temperature: 10–15 °C, wind 3–4 m/s). Collected water was measured volumetrically at the end of each event, and averaged.

Preliminary Testing

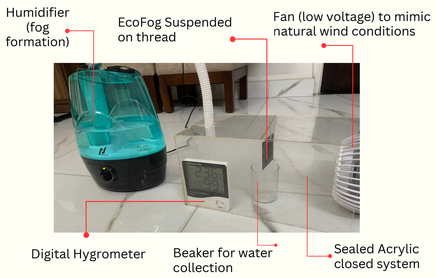


Fig. 11: Fog Chamber Set-up. *Photo taken by finalists, 2026.*

Results & Discussion

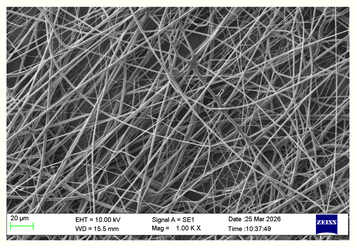


Fig. 12: FESEM of Crosslinked Electrospun Matrix. *Image taken by finalists, 2026.*

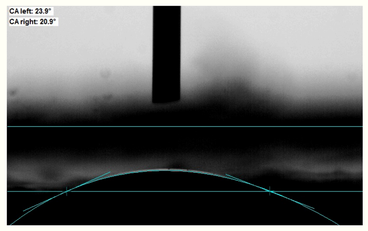


Fig 13: Contact Angle of Electrospun Matrix. *Image taken by finalists, 2026.*

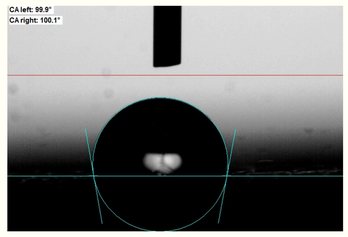


Fig. 14: Contact Angle of PLA Backbone. *Image taken by finalists, 2026.*

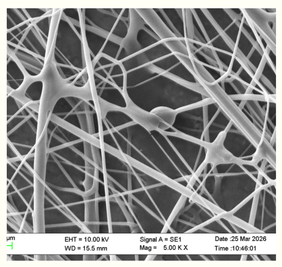


Fig. 15: FESEM of Base Matrix with 2 wt% TiO_2 and 1 wt% SiO_2. *Image taken by finalist, 2026.*

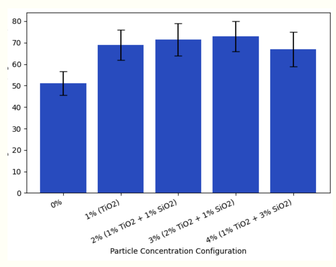


Fig. 16: Effect of Particle Concentration on Water Collection Rate. *Graph created by finalists using Matplotlib, 2026.*

At **1wt% TiO_2** and **1wt% SiO_2**, a high water collection rate was observed along with low nanoparticle agglomeration and bead formation.

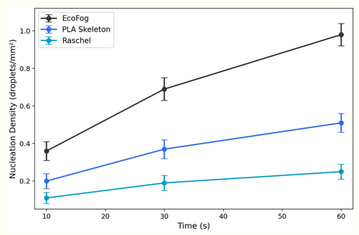


Fig. 17: Nucleation Density Over Time Across EcoFog, PLA Backbone, and Raschel Mesh. *Graph created by finalists using Matplotlib, 2026.*

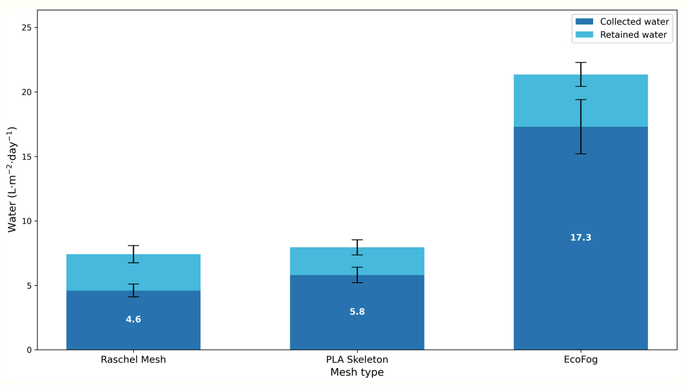


Fig. 18: FIELD TESTING RESULT: Water Collection (L m^{-2} day^{-1}) and Retained-Water Fraction on Mesh for EcoFog, PLA backbone, and Raschel Mesh. *Graph created by finalists using Matplotlib, 2026.*

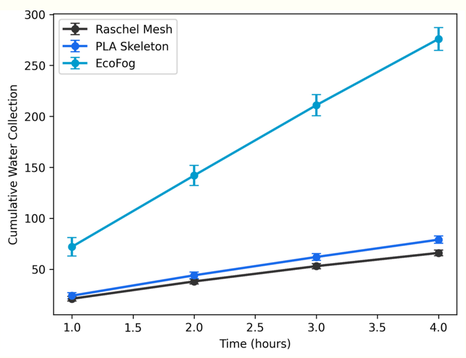


Fig. 19: Cumulative Water Collection (mg cm^{-2}) over 4 hours across EcoFog, PLA backbone, and Raschel Mesh. *Graph created by finalists using Matplotlib, 2026.*

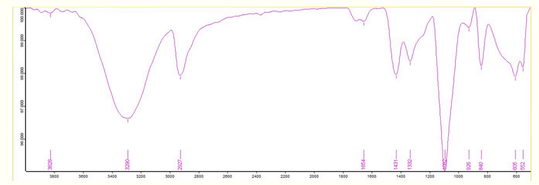


Fig. 20: FTIR spectrum of Crosslinked Electrospun Matrix. *Image taken by finalists, 2026.*

PVA O–H (3290 cm^{-1}), C–O (1092 cm^{-1}); new Ti–O (552, 605 cm^{-1}), Si–O–Si (1000cm^{-1}) confirm nanoparticle-polymer bonding post-crosslinking

Statistical Analysis

2D Surface Porosity	Water Collection Rate
• FESEM micrographs were analysed via ImageJ (Otsu thresholding, n = 4 images). • Mean **2D surface porosity** was 48.60 ± 3.89% (95% CI: 42.40-54.80%), exceeding literature **[8]**, confirming that the interfiber network remained open despite nanoparticle loading. • The **CV** of 8.01% confirms spatial homogeneity & reproducibility.	• **One-way ANOVA** shows strong mesh effect ($F(2,12)=124.15$, $p<0.001$). • **Tukey**: EcoFog significantly outperforms PLA and Raschel mesh; PLA and Raschel show no significant difference. • Large effect size ($\eta^2 \approx 0.95$). • The **CV** of 10-14% confirms consistent, reproducible performance across trials.

Conclusions

This study demonstrates that a system fabricated through **scalable techniques** and **widely-used materials** can serve as an effective water collector. This is possible through combining **high nucleation density** along with **faster coalescence** (through **hydrophilic** nanofibers/nanoparticles) and **faster drainage/reduced clogging** (through a **porous structure** built on a **hydrophobic** backbone).

Bench Impact: The developed mesh illustrates the possibility of an effective solution to water scarcity that not only maximizes water collection but also utilizes non-toxic, environmentally-friendly materials and scalable processes, facilitating large-scale deployment.

Field Impact: (1) The mesh meets the daily WHO minimum water requirements (50 L $person^{-1}$ day^{-1}) for 12 individuals per 40 m^2 system **[9]**; **(2)** It allows prevention of plastic persistence through the use of biodegradable materials.

Future Directions

- Optimize nanoparticle shapes.
- Utilize high-speed imaging to quantify nucleation, coalescence, pinning, and shedding kinetics.
- Develop a larger, modular prototype for scalable deployment through blanking of extruded sheets.
- Develop a coating to remove contaminants from smog.

References

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Facts and Graphics Citations:

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